HW 2

1.

(a)
Let A, B, C, D, E be the activity durations. The finish times in each replication are

$$f_A = A, \qquad f_B = B, \qquad f_C = A + C, \qquad f_D = \max(A, B) + D$$

$$T = \max(f_C,\, f_D) + E$$

Each replication i records three outputs: the completion time T_i, the deadline indicator $\mathbf{1}\{T_i > 22\}$, and the indicator $\mathbf{1}\{f_{C,i} > f_{D,i}\}$ that C finishes after D. Each activity has its own independent random-number stream. We create a `numpy.random.SeedSequence` and call `.spawn(5)` to make five independent generators, one per activity. Each generator samples only its own activity's distribution: Triangular(2, 4, 7) for A, Uniform(3, 6) for B, Exponential with mean 15 for C, Triangular(1, 3, 5) for D, and Uniform(2, 5) for E. The estimates are the sample means of the three outputs over n replications:

$$\bar{T} = \frac{1}{n}\sum_{i=1}^{n} T_i, \qquad \hat{p}_{22} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{T_i > 22\}, \qquad \hat{p}_{CD} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{f_{C,i} > f_{D,i}\}$$

With $n = 500{,}000$ replications (the same run as the last row of part d): - Estimated expected completion time: $\bar{T} \approx 23.28$ weeks - Estimated probability of missing the 22-week deadline: $\hat{p}_{22} \approx 0.3905$ - Estimated probability that C finishes after D: $\hat{p}_{CD} \approx 0.7873$

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(b)

Rep 1: A=4.875 B=4.427 C=3.996 D=1.956 E=3.833  T=12.705
Rep 2: A=3.320 B=4.802 C=1.135 D=3.917 E=3.159  T=11.878
Rep 3: A=4.693 B=3.735 C=3.216 D=3.105 E=4.533  T=12.442

(c)

[Name] [Simulated] [Theoretical Mean]
A 4.332652521726656 4.333333333333333
B 4.5016855766200745 4.5
C 14.950777075084364 15
D 2.9988973133064616 3
E 3.501325918026671 3.5

(a) (after 500,000 replications)

Estimated project completion time: 23.227056803437748
```

```
Estimated probability of missing the 22-week deadline: 0.388946
Estimated probability C finishes after D: 0.78645
```

For each output, the 95% confidence interval is

$$\bar{X} \pm t_{n-1,\,0.975}\,\frac{s}{\sqrt{n}}$$

where $\bar{X}$ and s are the sample mean and standard deviation of the replication outputs. For the two probabilities, the outputs are 0/1 indicators, so $\bar{X} = \hat{p}$ and $s = \sqrt{n\hat{p}(1-\hat{p})/(n-1)}$. For large n, $t_{n-1,\,0.975} \approx 1.96$.

n	$E[T]$ **est.**	**95% CI**	$P(T > 22)$ **est.**	**95% CI**	$P(C \text{ after } D)$ **est.**	**95% CI**
10	21.2729	(12.4423, 30.1035)	0.4000	(0.0306, 0.7694)	0.6000	(0.2306, 0.9694)
50	20.4760	(16.9460, 24.0060)	0.2800	(0.1511, 0.4089)	0.8000	(0.6852, 0.9148)
100	23.1427	(20.5614, 25.7241)	0.4100	(0.3119, 0.5081)	0.7400	(0.6525, 0.8275)
1,000	22.2696	(21.4467, 23.0924)	0.3700	(0.3400, 0.4000)	0.7740	(0.7480, 0.8000)
10,000	23.1845	(22.8942, 23.4748)	0.3788	(0.3693, 0.3883)	0.7859	(0.7779, 0.7939)
100,000	23.2534	(23.1626, 23.3442)	0.3911	(0.3881, 0.3941)	0.7876	(0.7851, 0.7902)
500,000	23.2782	(23.2375, 23.3189)	0.3905	(0.3892, 0.3919)	0.7873	(0.7861, 0.7884)

For small n, the probability intervals rely on a normal approximation that is rough. At $n = 10$ there are only about 4 or 6 "successes," so those intervals should be read as very approximate.

(e)
Interval widths. The half-width is proportional to $s/\sqrt{n}$, so it shrinks roughly like $1/\sqrt{n}$. Our results follow this closely. For $E[T]$, the full width falls from 17.66 at $n = 10$ to 0.081 at $n = 500{,}000$. Going from $n = 10{,}000$ to $100{,}000$ multiplies n by 10, and the width shrinks by a factor of about $3.2 \approx \sqrt{10}$. Going from $100{,}000$ to $500{,}000$, it shrinks by about $2.2 \approx \sqrt{5}$.

Must an interval narrow at every step? No. The width also depends on s (and on $\hat{p}(1-\hat{p})$ for the probabilities), which is itself a random estimate. If a larger run happens to produce a larger

s, the interval can get wider. For example, going from $n = 50$ to $n = 100$ only shrinks $1/\sqrt{n}$ by a factor of 0.71, so a run whose s is more than about 41% larger would give a wider interval. In our table every interval did narrow, because each step increases n by at least a factor of 2 and s was fairly stable, but this is not guaranteed.

Must the point estimates move steadily in one direction? No. Each estimate fluctuates randomly around its true value, and only the size of the fluctuations shrinks as n grows. Our estimates of $E[T]$ go 21.27, 20.48, 23.14, 22.27, 23.18, 23.25, 23.28: down, up, down, then up. $P(T > 22)$ goes 0.40, 0.28, 0.41, 0.37, 0.379, 0.391, 0.3905, and $P(C \text{ after } D)$ also moves both ways. The estimates settle down near 23.28, 0.39, and 0.787 for large n.

Coverage. A 95% interval is not guaranteed to contain the true value. Treating the $n = 500{,}000$ estimates as close to the truth, the $n = 1{,}000$ interval for $E[T]$, (21.45, 23.09), misses 23.28. The $n = 10{,}000$ interval for $P(T > 22)$, (0.3693, 0.3883), falls just short of 0.3905. Occasional misses like these are expected: about 1 in 20 intervals should miss.

2.

(a)

$$\begin{aligned} &\max 40x_A + 55x_B - (d_A - x_A)\cdot 15 - (d_B - x_B)\cdot 25 \\ &s.t.\ x_A + x_B \le 180, \quad 0 \le x_A \le d_A, \quad 0 \le x_B \le d_B \end{aligned}$$

We can simplify. Since each unit of B is worth \$80 and each unit of A is worth \$55, capacity goes to B first:

$$\begin{aligned} x_B^* &= \min(d_B, 180), \quad x_A^* = \min(d_A, 180 - x_B^*) \\ \pi^*(d_A, d_B) &= 55x_A^* + 80x_B^* - 15d_A - 25d_B \end{aligned}$$

(b) The covariance is $\mathrm{Cov}(D_A, D_B) = 0.4(25)(20) = 200$, so

$$\Sigma = \begin{pmatrix} 625 & 200 \\ 200 & 400 \end{pmatrix}, \qquad L = \begin{pmatrix} 25 & 0 \\ 8 & \sqrt{336} \end{pmatrix} \approx \begin{pmatrix} 25 & 0 \\ 8 & 18.33 \end{pmatrix}$$

where $\Sigma = LL^T$ is the Cholesky factorization. We draw $Z_1, Z_2 \sim N(0,1)$ independently from two separate random-number streams spawned with `numpy.random.SeedSequence()`. Then we set $D = \mu + LZ$:

$$D_A = 120 + 25Z_1, \qquad D_B = 100 + 8Z_1 + 18.33Z_2$$

This gives $\mathrm{Var}(D_B) = 64 + 336 = 400$ and $\mathrm{Cov}(D_A, D_B) = 25 \cdot 8 = 200$, as required. Profit is computed with $d_A = \max(D_A, 0)$ and $d_B = \max(D_B, 0)$. The exceedance event uses the original draws.

The pilot run check:

```
n=2000
mean: [120.04842155  99.65912825], std: [25.79955271 20.28754709],
correlation: 0.3978334502068665
```

These are close to the targets of (120, 100), (25, 20), and 0.4

(c)
Each replication gives a daily profit $Y_i = \pi^*(d_{A,i}, d_{B,i})$ and an indicator $I_i = \mathbf{1}\{D_{A,i} + D_{B,i} > 180\}$, for $i = 1, \ldots, n$.
Profit. The CI half-width is $h = 1.96\, s/\sqrt{n}$. To get $h \leq 50$, we need

$$n_{\text{profit}} \geq \left(\frac{1.96\, s}{50}\right)^2$$

Probability. With $\hat{p} = \frac{1}{n}\sum I_i$, the half-width is $h = 1.96\sqrt{\hat{p}(1-\hat{p})/n}$. To get $h \leq 0.01$, we need

$$n_{\text{prob}} \geq \left(\frac{1.96}{0.01}\right)^2 \hat{p}(1-\hat{p})$$

Pilot estimates ($n = 2000$): $\bar{Y} = 7930.11$, $s = 476.81$, $\hat{p} = 0.8555$.

$$n_{\text{profit}} \geq \left(\frac{1.96 \times 476.81}{50}\right)^2 = (18.691)^2 = 349.4 \;\Rightarrow\; 350$$

$$n_{\text{prob}} \geq 38{,}416 \times 0.8555 \times 0.1445 = 4748.98 \;\Rightarrow\; 4749$$

```
n=2000
ybar = 7930.107638053138, s = 476.81102544686263, p_hat = 0.8555
n_profit = 350, n_prob = 4749, n_req = 4749
```

- **number of replications required for 95% CI for profit within $50:** 350
- **number of replications required for 95% CI for $P(D_A + D_B > 180)$ within 0.01:** 4749
  Both outputs come from the same simulation run, so we use $n = \max(350, 4749) = 4749$.

(d)

```
n0=4749
ybar = 7948.27300720642, s = 465.23970478422297, p_hat = 0.8562421185372006
n_profit = 333, n_prob = 4729, n_req = 4729
```

Results with $n = 4749$:

Output	Point estimate	95% CI	Half-width	Goal	Met?
Expected daily profit	\$7948.27	[\$7935.04, \$7961.51]	\$13.23	≤ \$50	Yes
$P(D_A + D_B > 180)$	0.8562	[0.8463, 0.8662]	0.0100	≤ 0.01	Yes

Here $s = 465.24$, so the profit half-width is $1.96 \times 465.24/\sqrt{4749} = 13.23$. The probability half-width is $1.96\sqrt{0.8562 \times 0.1438/4749} = 0.00998$. Both intervals met their goals. Recomputing the requirement from these outputs gives $n_{\text{prob}} = 4729 \le 4749$, which confirms this. The profit interval is much tighter than required, because n was chosen for the probability goal. If an interval had missed its goal, we would recompute the requirement from the new estimates. For example, $n_{\text{new}} = (1.96/0.01)^2\,\hat{p}(1-\hat{p})$, or equivalently $n_{\text{new}} = n\,(h_{\text{obs}}/h_{\text{target}})^2$, rounded up. We would then run the additional $n_{\text{new}} - n$ replications, possibly with a small safety margin.

(e)

```
S ~ N(220, 1425.0), z = -1.0596, P = 0.8553
p_hat = 0.8562, 95% CI = [0.8463, 0.8662], half-width = 0.0100
theory inside CI: True
```

Let $S = D_A + D_B$. A sum of jointly normal variables is normal, with

$$E[S] = 120 + 100 = 220$$

$$\mathrm{Var}(S) = 25^2 + 20^2 + 2(0.4)(25)(20) = 625 + 400 + 400 = 1425, \qquad \sigma_S = \sqrt{1425} = 37.75$$

So $S \sim N(220,\, 1425)$. The standardized threshold is

$$z = \frac{180 - 220}{37.75} = -1.0596$$

$$P(S > 180) = 1 - \Phi(-1.0596) = \Phi(1.0596) \approx 0.8553$$

Comparison. The simulation estimate is $\hat{p} = 0.8562$ with 95% CI [0.8463, 0.8662]. The theoretical value 0.8553 is contained in the interval, and it differs from the estimate by only about 0.0009, well within the half-width of 0.0100. This is consistent with the generator correctly reproducing the specified means, variances, and correlation. The covariance term matters. Without it, $\mathrm{Var}(S) = 1025$ and $P(S > 180) \approx 0.894$, which would fall outside the interval. Clipping negative demands does not affect this comparison, because the exceedance event uses the original normal draws, and negative draws are extremely rare anyway, on the order of 10^{-6}.